Minh Nguyen

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Nationality: Australian, Vietnamese
Languages: English (fluent), Vietnamese (native)
Location: Brisbane, QLD, Australia

Current position

Specialist Data Scientist, BHP

Skills

Programming: Python, R, SQL, SAS, MATLAB, LaTeX

Machine Learning & Data Science: Scikit-Learn, XGBoost, LightGBM, PyTorch, TensorFlow, Keras, NumPy, Pandas, Matplotlib, Streamlit

Business Intelligence & Visualisation: Power BI, SSRS, DAX

Cloud & Data Platforms: GCP, Vertex AI, Snowflake, AWS Sagemaker, SSMS, HEX

AI Tools & NLP: LlamaIndex framework, Claude Code, Model Context Protocol servers, Microsoft Foundry (Azure AI Foundry)

Geospatial & Location Intelligence: GeoPandas, MapInfo (GIS software)

Mathematical Foundations: statistical modelling, probabilistic models, stochastic processes, numerical linear algebra, computational mathematics

Development Tools: Azure DevOps, Bitbucket, Jupyter Notebook, Kiro IDE, VS Code, Git, GitHub, Jira, Confluence

Certificates

Microsoft Generative AI Engineering Professional Certificate , Coursera

Work experience

Aug 2026 – **Specialist Data Scientist (Contractor), BHP**
Duties:

- Support the Value Engineering team in assessing asset performance across Copper South Australia (CuSA) production operations
- Identify and close performance gaps through data-driven analysis of production processes
- Apply hypothesis testing on key operational strategies to identify key performance drivers (KPDs) influencing production performance, informing potential uplift opportunities
- Collaborate closely with the System Dynamics team, Product Owners, and Scrum Masters to test simulation hypotheses and validate modelling assumptions within an agile delivery framework
- Contribute to shared code repositories across multiple projects within the same asset-focused portfolio, supporting consistency and reuse of analytical approaches

Sep 2024 – Aug 2026 **Data Scientist, Sedgwick**
Duties:

- Led design and delivery of a production machine learning system for catastrophe claims forecasting, shifting internal operations from a reactive process to proactive, day-one visibility of claim volumes during live events - improving resource allocation efficiency and operational readiness during catastrophe response
- Applied a combination of machine learning and actuarial methods to significantly improve early forecasting accuracy, strengthening confidence in claims estimates across multiple catastrophe types
- Liaised directly with insurer and internal stakeholders to identify opportunities for machine learning and data science to uplift analytical capability, translating business requirements into technical solutions
- Developed automated geographic detection of catastrophe-affected areas using weather data, replacing manual assessment with a faster, systematic process, achieving 95%+ recall
- Built and maintained dashboards and reporting for internal and client-facing use, supporting both business-as-usual and live catastrophe event decision-making
- Automated recurring data and analysis workflows, reducing manual effort and improving consistency across reporting processes
- Presented the completed system's architecture, validation approach, and business impact to CEO and COO on delivery
- Provided general technical guidance and mentorship to junior data scientists and analysts, supporting them across a range of topics as needed, and performed ad-hoc analysis on request

Nov 2021 – Sep 2024

Pricing Analyst - Motor Insurance, Auto & General Insurance Holdings Pty Ltd
Duties:

- Developed and maintained generalised linear models (GLMs) for motor insurance pricing, incorporating geospatial variables, driving behaviour metrics, and vehicle characteristics to optimise risk-based premium accuracy
- Performed exploratory data analysis using LightGBM models and SHAP values to identify key feature importances and interaction effects, informing feature engineering and variable selection for production GLMs
- Designed and analysed A/B tests to evaluate pricing strategy changes, measuring impact on conversion rates, portfolio profitability, and customer retention across different market segments
- Built automated data pipelines using Python and SQL for extracting, transforming, and validating large-scale insurance data from multiple sources, reducing manual processing time and improving data quality
- Conducted rate change analysis and financial forecasting to support business planning, presenting recommendations to senior leadership on pricing adjustments and market positioning

Feb 2021 – Nov 2021

Teaching Assistant, University of Queensland
Courses:

- STAT1201 *Analysis of Scientific Data*
- COSC2500 *Numerical Methods in Computational Science*
- STAT2004 *Statistical Modelling & Analysis*
- STAT3500 *Problems & Applications in Modern Statistics*

Duties:

- Conducting tutorials and practicals
- Providing technical and IT support for online and on-campus lectures
- Offering academic support to first and second-year students
- Marking assessments and assignments

Education

(MDataSc) Master of Data Science, University of Queensland
Ongoing.

(BMath) Bachelor of Mathematics with Statistics major, University of Queensland
GPA 6.6/7. Completed.

Scholarships & awards

2021 UQ School of Maths and Physics Summer Industry Experience Award (approximately $3000), University of Queensland

2021 Dean's Commendation for Academic Excellence, University of Queensland

2020 Dean's Commendation for Academic Excellence, University of Queensland

2018 Dean's Commendation for Academic Excellence, University of Queensland

2018 – 2020 UQ Link Scholarship ($9000), University of Queensland

Conferences, talks & workshops

2025 – 2026 **Attendee at CDAO Brisbane Conference (link)**
Participated in discussions regarding timely and trending insights on governance, generative AI, machine learning, and data leadership strategies.

2022 **Participant in CPD151, Machine Learning with Python Workshop, hosted by the Statistical Society of Australia, Online (link)**
Engaged in a two-day intensive workshop tailored for data scientists, focusing on the construction of machine learning models within production-oriented workflow frameworks.

2022 **Invited Presenter at the SMP Summer Industry Experience Showcase 2021–2022, University of Queensland**
Selected to present my summer industrial project during a showcasing event organised by the University of Queensland's School of Mathematics and Physics. Presented to an audience comprising academic staff, students, and industry collaborators.

2019 **Attendee at BAM2019, Biarri Applied Mathematics Brisbane Conference (link)**
Participated in a multidisciplinary conference centred on mathematical modelling, statistical analyses, optimisation methods, and machine learning applications aimed at elevating business intelligence and industrial automation.

Affiliations

2021 – 2023 Student Member of the Statistical Society of Australia (SSA) (link)

2020 – 2023 Member of the University of Queensland's Mathematics Students Society (MSS) (link)

2024 Member of the Women in Technology Organisation (WiT) (link)